

Ayers AskDocs Phase 5: Composite vs Quality-Only Reward

Composite = mean(normalized quality, normalized empathy). Quality-only excludes empathy. Both use human healthcare-professional labels; no LLM judge scoring.

Dataset: 195 paired physician vs logged ChatGPT responses | Bootstrap: 100 | Seed: 0

Full estimator table

Reward	Config	Estimator	V_b	V_agent	Effect	V_hat	Bias	RMSE	95% CI	Cov
Composite	BGE-M3	DM	0.4254	0.7233	+0.2979	0.5537	-0.1696	0.1832	[0.502, 0.593]	no
Composite	BGE-M3	MIPS	0.4254	0.7233	+0.2979	0.6193	-0.1040	0.1134	[0.555, 0.696]	no
Composite	BGE-M3	OffCEM	0.4254	0.7233	+0.2979	0.5551	-0.1682	0.1826	[0.503, 0.593]	no
Composite	BGE-M3 + learned concat-PCA	DM	0.4254	0.7233	+0.2979	0.5593	-0.1640	0.1611	[0.531, 0.597]	no
Composite	BGE-M3 + learned concat-PCA	MIPS	0.4254	0.7233	+0.2979	0.6464	-0.0769	0.0857	[0.588, 0.716]	no
Composite	BGE-M3 + learned concat-PCA	OffCEM	0.4254	0.7233	+0.2979	0.5607	-0.1626	0.1603	[0.532, 0.597]	no
Quality-only	BGE-M3	DM	0.5641	0.7829	+0.2188	0.6914	-0.0915	0.1102	[0.635, 0.720]	no
Quality-only	BGE-M3	MIPS	0.5641	0.7829	+0.2188	0.7304	-0.0526	0.0612	[0.679, 0.776]	no
Quality-only	BGE-M3	OffCEM	0.5641	0.7829	+0.2188	0.6926	-0.0903	0.1098	[0.635, 0.720]	no
Quality-only	BGE-M3 + learned concat-PCA	DM	0.5641	0.7829	+0.2188	0.6983	-0.0846	0.0857	[0.668, 0.727]	no
Quality-only	BGE-M3 + learned concat-PCA	MIPS	0.5641	0.7829	+0.2188	0.7587	-0.0242	0.0363	[0.706, 0.801]	yes
Quality-only	BGE-M3 + learned concat-PCA	OffCEM	0.5641	0.7829	+0.2188	0.6998	-0.0831	0.0848	[0.669, 0.727]	no

Direct MIPS comparison

Reward	Config	V_agent	MIPS V_hat	Bias	RMSE	Cov
Composite	BGE-M3	0.7233	0.6193	-0.1040	0.1134	no
Composite	BGE-M3 + learned concat-PCA	0.7233	0.6464	-0.0769	0.0857	no
Quality-only	BGE-M3	0.7829	0.7304	-0.0526	0.0612	no
Quality-only	BGE-M3 + learned concat-PCA	0.7829	0.7587	-0.0242	0.0363	yes

Takeaways

- Composite ground truth: V_b=0.4254, V_agent=0.7233, effect=+0.2979.
- Quality-only ground truth: V_b=0.5641, V_agent=0.7829, effect=+0.2188.
- Composite MIPS RMSE improves 0.1134 -> 0.0857 with learned concat-PCA.
- Quality-only MIPS RMSE improves 0.0612 -> 0.0363; this is the only covering CI.